

**Born too soon**

Read the World Health Organisation's global action report on premature deliveries. <https://digit.in/mar20-49>

**No wires more cuddles**

Soft, flexible wireless sensors for monitoring premature babies allow their parents to give more cuddles. <https://digit.in/mar20-50>

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normal pregnancy in humans lasts for about 40 weeks. During this time, the infant gradually

develops within the womb, protected by a fluid-filled structure known as the amniotic sac. If a baby is born before 40 weeks, it is called premature, and a baby born after 40 weeks is called postmature. Premature babies can suffer from a number of health problems throughout their lives. A baby is considered to be premature if it is born before 37 weeks of the pregnancy. While most premature babies are born between 34 to 36 weeks into the pregnancy, some are born extremely premature, which is considered to be at 25 weeks or before. Babies born before 23 weeks are not considered viable by doctors, and are not resuscitated. Every year, one million premature babies around the world do not survive. A premature birth is the biggest cause for newborn deaths. Even if the baby survives, they may face health problems throughout their lives, as the normal development of the infant is affected. These include hearing loss, abnormally large or inflamed lungs, asthma, hearing loss, delayed teeth growth, mental health problems and neurological disorders. The earlier a child is born, the more problems it faces. There are some identified known factors that increase the cause of a premature birth. These include a previous pregnancy with twins, having a pregnancy soon after childbirth, a family history of premature births, diabetes and inadequate healthcare to the mother. The most reliable indicator that a birth will be preterm, is that the mother has previously given birth to a preterm baby. Air pollution is also known to increase the chances of a premature childbirth, and all around the world, the number of premature childbirth is increasing. However, the cause of over 50 percent of premature child births are unknown. While these might seem like a lot of statistics, there is a story behind each and every one of those numbers. While 9 out of 10



PREDICTING PREMATURE CHILDBIRTH

About 18% of all deliveries worldwide are premature and there is not yet a reliable way to predict such births.

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premature babies go on to live mostly normal lives, if premature births can be accurately predicted, then doctors can take measures to extend the gestation period, to prevent health problems later in life to the individual. However, the efforts by scientists and healthcare

workers to predict premature births have so far been very frustrating. The current tests for predicting premature births work best on those mothers who have already had a premature birth, and give an accurate prediction only 20 percent of the time. Additionally, scientists are also struggling to accurately predict the gestation period of any pregnancy. Once the biological mechanisms for premature childbirth